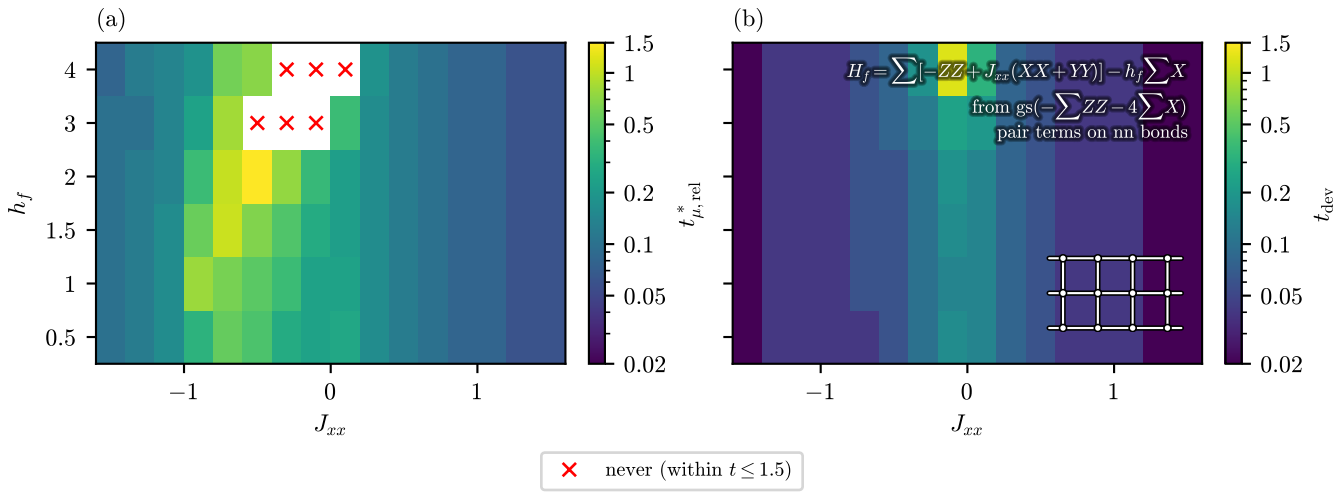

(a)
(b)
h_f
J_{xx}
$t^*_{\mu,\mathrm{rel}}$
t_{dev}
$H_f = \sum[-ZZ + J_{xx}(XX+YY)] - h_f \sum X$
from gs$(-\sum ZZ - 4\sum X)$
pair terms on nn bonds
never (within $t \leq 1.5$)